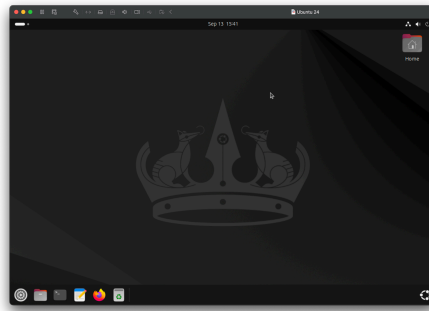


Lab 3

Stephanie Salgado

Using a Linux VM



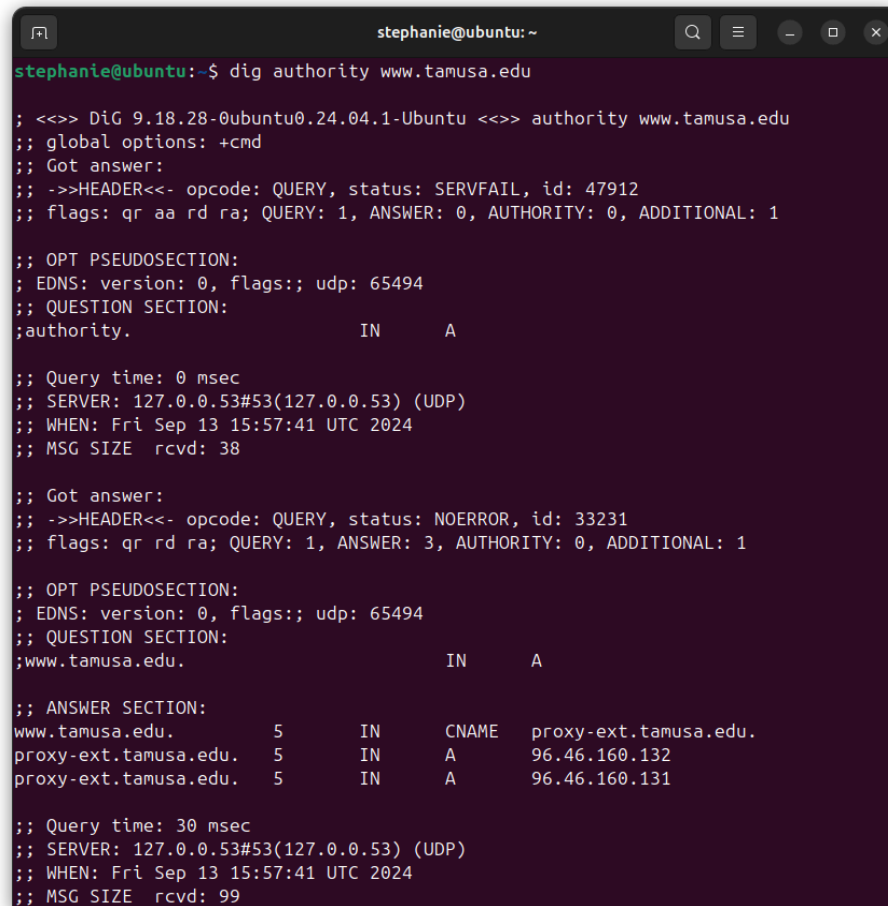
Since we needed Linux for this lab, I used VMware Fusion on my Mac to add and configure an Ubuntu 24 VM.

The “dig” command

```
stephanie@ubuntu: ~  
DIG(1)                                BIND 9                                DIG(1)  
  
NAME  
    dig - DNS lookup utility  
  
SYNOPSIS  
    dig [@server] [-b address] [-c class] [-f filename] [-k filename] [-m]  
    [-p port#] [-q name] [-t type] [-v] [-x addr] [-y [hmac:]name:key] [  
    [-4] | [-6] ] [name] [type] [class] [queryopt...]  
  
    dig [-h]  
  
    dig [global-queryopt...] [query...]  
  
DESCRIPTION  
    dig is a flexible tool for interrogating DNS name servers. It performs  
    DNS lookups and displays the answers that are returned from the name  
    server(s) that were queried. Most DNS administrators use dig to trou-  
    bleshoot DNS problems because of its flexibility, ease of use, and  
    clarity of output. Other lookup tools tend to have less functionality  
    than dig.  
  
    Although dig is normally used with command-line arguments, it also has  
    Manual page dig(1) line 1 (press h for help or q to quit)
```

The **d**omain **i**nformation **g**roper command is used to gather DNS information.

Using “dig authority” on tamusa

A terminal window titled 'stephanie@ubuntu: ~' showing the output of the command 'dig authority www.tamusa.edu'. The output is split into two sections. The first section shows a 'SERVFAIL' status for the query to the local DNS resolver (127.0.0.53). The second section shows a 'NOERROR' status for the query to the authoritative server (127.0.0.53#53). The answer section of the second query shows that 'www.tamusa.edu' is a CNAME record pointing to 'proxy-ext.tamusa.edu', which is an 'A' record with two IP addresses: 96.46.160.132 and 96.46.160.131. The TTL for these records is 5 seconds.

```
stephanie@ubuntu:~$ dig authority www.tamusa.edu

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> authority www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 47912
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;authority.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 15:57:41 UTC 2024
;; MSG SIZE rcvd: 38

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33231
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

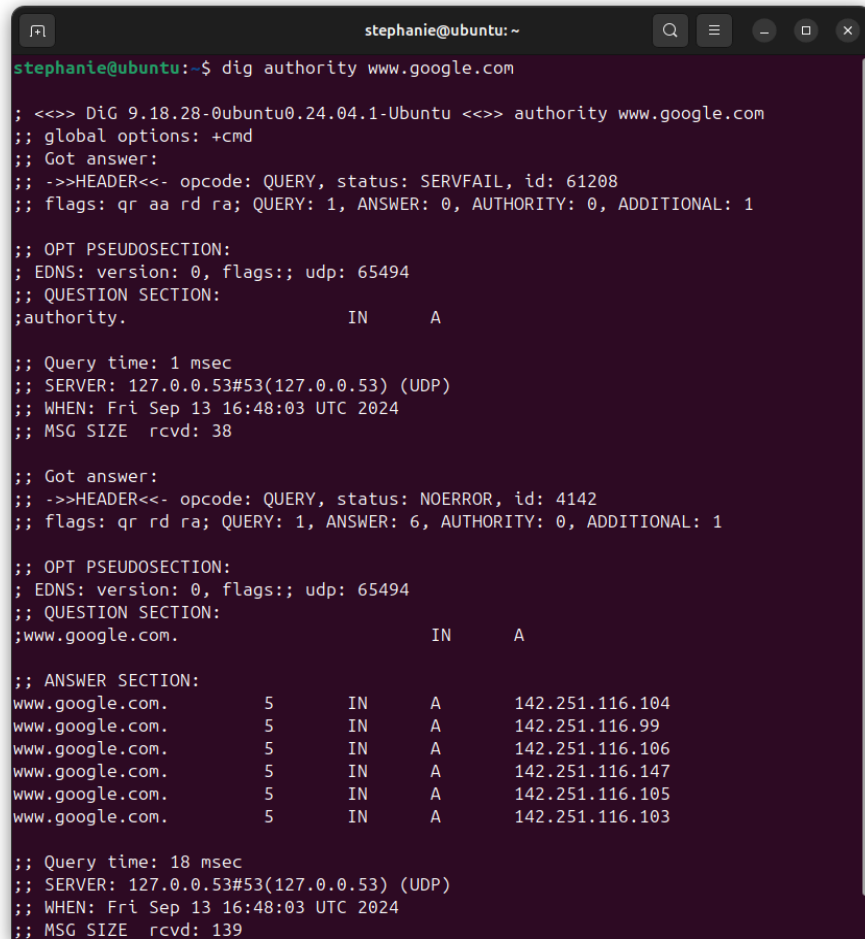
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;www.tamusa.edu.           IN      A

;; ANSWER SECTION:
www.tamusa.edu.            5       IN      CNAME   proxy-ext.tamusa.edu.
proxy-ext.tamusa.edu.      5       IN      A       96.46.160.132
proxy-ext.tamusa.edu.      5       IN      A       96.46.160.131

;; Query time: 30 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 15:57:41 UTC 2024
;; MSG SIZE rcvd: 99
```

The answer section is what provides the information we queried. In this case, the first query had a status of “SERVFAIL” indicating there was a failure resolving the DNS query. The query was made to “127.0.0.53”, the DNS resolver. The second query had a status of “NOERROR”, meaning the DNS query was successful. The answer section reflects this. The first column in this section corresponds to the domain names, notice the 4th column, it says “CNAME” for “www.tamusa.edu”, this shows it’s an alias pointing to “proxy-ext.tamusa.edu”. The “A” for both is the “Address” record, which resolves to 2 IPs. The “5” signifies the TTL (in seconds), which defines how long the DNS record is cached. The “IN” simply indicates the DNS record class is Internet.

Using “dig authority” on google



```
stephanie@ubuntu:~$ dig authority www.google.com

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> authority www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 61208
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;authority.                IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 16:48:03 UTC 2024
;; MSG SIZE rcvd: 38

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 4142
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com.           IN      A

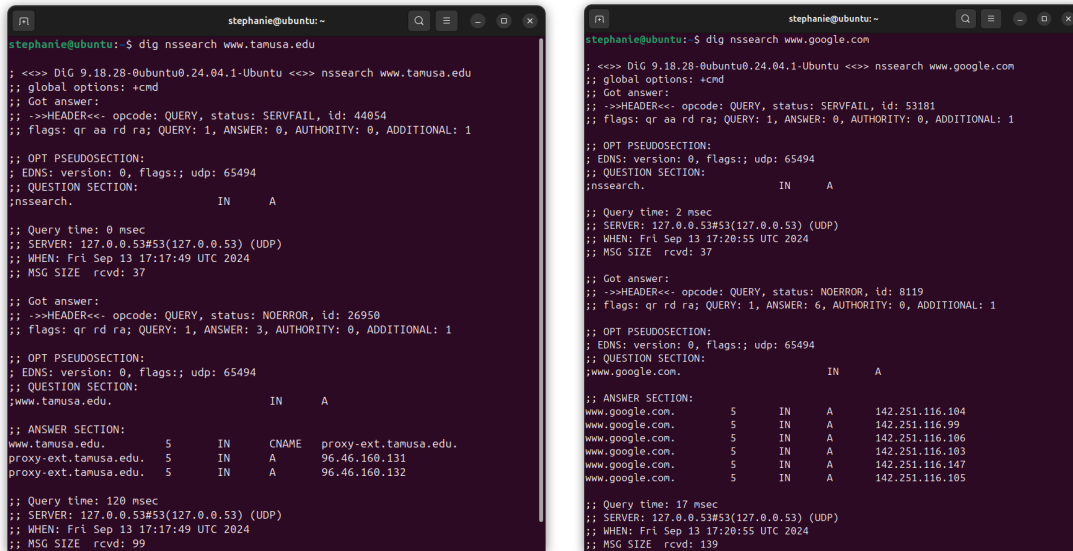
;; ANSWER SECTION:
www.google.com.           5      IN      A       142.251.116.104
www.google.com.           5      IN      A       142.251.116.99
www.google.com.           5      IN      A       142.251.116.106
www.google.com.           5      IN      A       142.251.116.147
www.google.com.           5      IN      A       142.251.116.105
www.google.com.           5      IN      A       142.251.116.103

;; Query time: 18 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 16:48:03 UTC 2024
;; MSG SIZE rcvd: 139
```

The first query had the same status as when I used it on “tamura”. The main difference here is that “www.google.com” has multiple “A” records (IPv4 addresses), which is normal with high traffic websites for load balancing. Meaning anytime you query the site, the DNS can return any of the IPs. Additionally, the query time for google was quicker, at 18ms, when compared to 30ms for “tamura”.

I noticed that if the first query was successful for both “tamura” and “google”, there would’ve been an “SOA”, “Start of Authority”, record which would display “a.root-servers.net. nstld.verisign-grs.com.” under the authority section.

Using “dig nssearch” on tamusa, google and facebook



```
stephanie@ubuntu:~$ dig nssearch www.tamusa.edu
;; <<>> DIG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> nssearch www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 44054
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 65494
;; QUESTION SECTION:
;; nssearch.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:17:49 UTC 2024
;; MSG SIZE rcvd: 37

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26950
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 65494
;; QUESTION SECTION:
;; www.tamusa.edu.          IN      A

;; ANSWER SECTION:
www.tamusa.edu.      5       IN      CNAME   proxy-ext.tamusa.edu.
proxy-ext.tamusa.edu. 5       IN      A       96.46.160.131
proxy-ext.tamusa.edu. 5       IN      A       96.46.160.132

;; Query time: 120 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:17:49 UTC 2024
;; MSG SIZE rcvd: 99

stephanie@ubuntu:~$ dig nssearch www.google.com
;; <<>> DIG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> nssearch www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 53181
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 65494
;; QUESTION SECTION:
;; nssearch.                IN      A

;; Query time: 2 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:20:55 UTC 2024
;; MSG SIZE rcvd: 37

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8119
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 65494
;; QUESTION SECTION:
;; www.google.com.          IN      A

;; ANSWER SECTION:
www.google.com.      5       IN      A       142.251.116.104
www.google.com.      5       IN      A       142.251.116.99
www.google.com.      5       IN      A       142.251.116.106
www.google.com.      5       IN      A       142.251.116.103
www.google.com.      5       IN      A       142.251.116.147
www.google.com.      5       IN      A       142.251.116.105

;; Query time: 17 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:20:55 UTC 2024
;; MSG SIZE rcvd: 139
```

I got the “SERVFAIL” status when trying “dig nssearch”. From the sample, I know the command would have given similar SOA information for each.

Trying on Mac

For “www.tamusa.edu”:

```
;; QUESTION SECTION:
;nssearch.                IN      A

;; AUTHORITY SECTION:
.                86399   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091300 1800 900 604800 86400
```

For “www.google.com”:

```
;; QUESTION SECTION:
;nssearch.                IN      A

;; AUTHORITY SECTION:
.                86390   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091300 1800 900 604800 86400
```

For “www.facebook.com”:

```
;; QUESTION SECTION:
;nssearch.                IN      A

;; AUTHORITY SECTION:
.                86395   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

Using “dig additional” on facebook and tamusa

```
stephanie@ubuntu:~$ dig additional www.tamusa.edu

;<> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <> additional www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 2531
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
```

Notice I got the “SERVFAIL” status again. I decided to continue on Mac.

```
dig additional www.tamusa.edu

;<> DiG 9.18.6 <> additional www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 7181
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 512
;; QUESTION SECTION:
;additional. IN A
;; AUTHORITY SECTION:
a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
;; Query time: 37 msec
;; SERVER: 8.8.8.853(8.8.8.8)
;; WHEN: Fri Sep 13 12:19:18 CDT 2024
;; MSG SIZE rcvd: 114
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 5423
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 512
;; QUESTION SECTION:
;www.tamusa.edu. IN A
;; ANSWER SECTION:
www.tamusa.edu. 3167 IN CNAME proxy-ext.tamusa.edu.
proxy-ext.tamusa.edu. 125 IN A 96.46.168.132
proxy-ext.tamusa.edu. 125 IN A 96.46.168.131
;; Query time: 14 msec
;; SERVER: 8.8.8.853(8.8.8.8)
;; WHEN: Fri Sep 13 12:19:18 CDT 2024
;; MSG SIZE rcvd: 99
```

```
dig additional www.facebook.com

;<> DiG 9.18.6 <> additional www.facebook.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 3869
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 512
;; QUESTION SECTION:
;additional. IN A
;; AUTHORITY SECTION:
a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
;; Query time: 22 msec
;; SERVER: 8.8.8.853(8.8.8.8)
;; WHEN: Fri Sep 13 12:19:18 CDT 2024
;; MSG SIZE rcvd: 114
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 53839
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 512
;; QUESTION SECTION:
;www.facebook.com. IN A
;; ANSWER SECTION:
www.facebook.com. 2777 IN CNAME star-mini.cdn.facebook.com.
star-mini.cdn.facebook.com. 33 IN A 157.248.25.35
;; Query time: 28 msec
;; SERVER: 8.8.8.853(8.8.8.8)
;; WHEN: Fri Sep 13 12:19:18 CDT 2024
;; MSG SIZE rcvd: 98
```

The first queries returned the same SOA records. The only major difference I can spot is in the TTLs. For some reason, they are larger with this kind of query.

Using “dig nsid” on tamusa and facebook

For “www.tamusa.edu”:

```
;; QUESTION SECTION:
;nsid. IN A

;; AUTHORITY SECTION:
a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

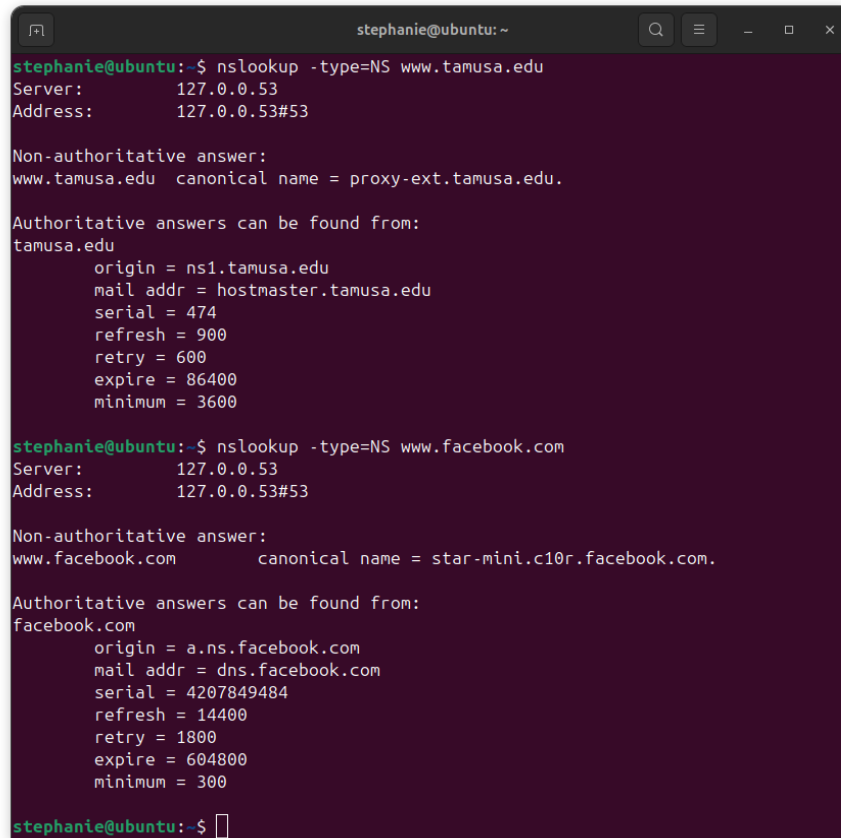
For “www.facebook.com”:

```
;; QUESTION SECTION:
;nsid. IN A

;; AUTHORITY SECTION:
a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

Again, the queries return the same SAO records and show no further changes.

Using “nslookup -type=NS” on tamusa and facebook

A terminal window titled 'stephanie@ubuntu: ~' with search, menu, and window control icons. It shows two nslookup commands. The first is for 'www.tamusa.edu', showing the server 127.0.0.53, a non-authoritative answer pointing to proxy-ext.tamusa.edu, and authoritative DNS details for tamusa.edu. The second is for 'www.facebook.com', showing the same server, a non-authoritative answer pointing to star-mini.c10r.facebook.com, and authoritative DNS details for facebook.com.

```
stephanie@ubuntu:~$ nslookup -type=NS www.tamusa.edu
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
www.tamusa.edu canonical name = proxy-ext.tamusa.edu.

Authoritative answers can be found from:
tamusa.edu
    origin = ns1.tamusa.edu
    mail addr = hostmaster.tamusa.edu
    serial = 474
    refresh = 900
    retry = 600
    expire = 86400
    minimum = 3600

stephanie@ubuntu:~$ nslookup -type=NS www.facebook.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.

Authoritative answers can be found from:
facebook.com
    origin = a.ns.facebook.com
    mail addr = dns.facebook.com
    serial = 4207849484
    refresh = 14400
    retry = 1800
    expire = 604800
    minimum = 300

stephanie@ubuntu:~$
```

The “nslookup -type=NS” command outputs look simpler and easier to read. It returns the appropriate “127.0.0.53” DNS resolver server/address. This is because the command uses the “-type” flag with the value of “NS”, which queries for NS (Name Server) records specifically.

For “tamusa”, it displays the same “CNAME” information and shows where it points. It provides further information under the “Authoritative answers can be found from:” section.

For “facebook”, it displays the “CNAME” information which is pointing to “star-mini.c10r.facebook.com”. It also provides all the “Authoritative answers can be found from:” section detailing information like origin, mail addr, etc,...

Using “nslookup” on tamusa and facebook

A screenshot of a terminal window titled 'stephanie@ubuntu: ~'. The terminal shows two nslookup commands and their outputs. The first command is 'nslookup www.tamusa.edu', which returns the server address 127.0.0.53 and a non-authoritative answer for www.tamusa.edu, showing its canonical name as proxy-ext.tamusa.edu and two IP addresses: 96.46.160.131 and 96.46.160.132. The second command is 'nslookup www.facebook.com', which returns the same server address and a non-authoritative answer for www.facebook.com, showing its canonical name as star-mini.c10r.facebook.com and two addresses: 31.13.89.35 and an IPv6 address 2a03:2880:f162:81:face:b00c:0:25de.

```
stephanie@ubuntu: ~  
stephanie@ubuntu:~$ nslookup www.tamusa.edu  
Server:      127.0.0.53  
Address:     127.0.0.53#53  
  
Non-authoritative answer:  
www.tamusa.edu canonical name = proxy-ext.tamusa.edu.  
Name:   proxy-ext.tamusa.edu  
Address: 96.46.160.131  
Name:   proxy-ext.tamusa.edu  
Address: 96.46.160.132  
  
stephanie@ubuntu:~$ nslookup www.facebook.com  
Server:      127.0.0.53  
Address:     127.0.0.53#53  
  
Non-authoritative answer:  
www.facebook.com canonical name = star-mini.c10r.facebook.com.  
Name:   star-mini.c10r.facebook.com  
Address: 31.13.89.35  
Name:   star-mini.c10r.facebook.com  
Address: 2a03:2880:f162:81:face:b00c:0:25de  
  
stephanie@ubuntu:~$
```

Finally, the “nslookup” commands provide a similar view. Without the “-type” tag, it returns the default “Address” records. It can be noted that this time, the command provides the “Non-authoritative answer:”. It still displays the canonical names and where they point for both. However, this time, the addresses are also shown, one of them for facebook is in IPv6.

Summary

In this lab, I got to use the “dig” and “nslookup” commands to learn more about DNS. I encountered a problem with the “dig authority” and “dig nssearch” commands having a “SERVFAIL” status each time I ran them. I found success when running them on Mac. I believe it was likely due to the systems having different versions of “dig”. I had no problems when running the “nslookup” commands in the VM. I liked “nslookup” because it was simple. From this lab, I was able to learn about “canonical names”, and different types of DNS records like “Start of Authority”, “Address” and “Name Server”.